1. Operators in Quantum mechanics

Physical Quantities	Operators	Symbol
Position	$\boldsymbol{x}$	$\hat{X}$
Linear Momentum (x-component)	$-i\hbar\frac{d}{dx}$	$\widehat{P_x}$
Kinetic Energy	$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$	$\hat{K}$
Potential Energy	$V(x)$	$\hat{V}$
Total Energy (free particle)	$i\hbar\frac{d}{dt}$	$\hat{E}$
Total Mechanical Energy (also known as Hamiltonian Operator)	$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+V(x)$	$\hat{H}$

2. Expectation values of measurable quantities

(i) Expectation value of position:

Let us consider a particle is described by a wave function Ψ, the average value of the position x of the particle, is given by expectation value

$$\langle x\rangle=\frac{\int_{-\infty}^{\infty}\Psi^*x\Psi dx}{\int_{-\infty}^{\infty}\Psi^*\Psi\, dx}=\frac{\int_{-\infty}^{\infty}\Psi^*x\Psi dx}{\int_{-\infty}^{\infty}|\Psi|^2\, dx}$$

If the function is normalized, i.e. $\int_{-\infty}^{\infty}\Psi^*\Psi\, dx=1$

Then $\langle x\rangle=\int_{-\infty}^{\infty}\Psi^*\hat{X}\Psi dx=\int_{-\infty}^{\infty}\Psi^*x\Psi dx=\int_{-\infty}^{\infty}x\Psi^*\Psi dx=\int_{-\infty}^{\infty}x|\Psi|^2\, dx$

(ii) Expectation value of momentum:

Expectation value of the momentum is defined by

$$\langle p\rangle=\int_{-\infty}^{\infty}\Psi^*\hat{p}\Psi dx$$

$$\Rightarrow\langle p\rangle=\int_{-\infty}^{\infty}\Psi^*\left(-i\hbar\frac{d}{dx}\right)\Psi dx$$

$$\Rightarrow\langle p\rangle=-i\hbar\int_{-\infty}^{\infty}\Psi^*\left(\frac{d\Psi}{dx}\right)dx$$

(iii) Expectation value of energy of the free particle:

Expectation value of the Energy is defined by

$$\langle E\rangle=\int_{-\infty}^{\infty}\Psi^*\hat{E}\Psi dx$$

$$\Rightarrow\langle E\rangle=\int_{-\infty}^{\infty}\Psi^*\left(i\hbar\frac{d}{dt}\right)\Psi dx$$

$$\Rightarrow\langle E\rangle=i\hbar\int_{-\infty}^{\infty}\Psi^*\left(\frac{d\Psi}{dt}\right)dx$$

(iv) Expectation value of kinetic energy of the particle:

Expectation value of the kinetic energy is defined by

$$\langle \boldsymbol{K} \rangle = \int_{-\infty}^{\infty} \boldsymbol{\Psi^* \hat{K} \Psi dx}$$

$$\Rightarrow \langle \boldsymbol{K} \rangle = \int_{-\infty}^{\infty} \boldsymbol{\Psi^*} \left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} \right) \boldsymbol{\Psi dx}$$

$$\Rightarrow \langle \boldsymbol{K} \rangle = -\frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \boldsymbol{\Psi^*} \left(\frac{d^2\Psi}{dx^2} \right) \boldsymbol{dx}$$

(v) Expectation value of potential energy of the particle:

Expectation value of the potential energy is defined by

$$\langle \boldsymbol{V} \rangle = \int_{-\infty}^{\infty} \boldsymbol{\Psi^* \hat{V} \Psi dx}$$

$$\Rightarrow \langle \boldsymbol{V} \rangle = \int_{-\infty}^{\infty} \boldsymbol{\Psi^* V(x) \Psi dx}$$

3. Operators and Eigen Values:

Certain dynamical variables ($suppose\ G$) are restricted to a discrete values(G_n), or in other words G be quantized. If the system is defined by wave function Ψ_n is such that:

$$\boldsymbol{\hat{G}\Psi_n = G_n\Psi_n}$$

Where,$\hat{G}$ is the operator corresponds to the physical quantity G, of which each discrete values G_n, is a real number.

Example: Eigen value of a total energy Operator (Hamiltonian Operator)

Suppose the Eigen values of a total energy operator $\hat{\mathbf{H}}$ are given by discrete values(E_n), and if the system is defined by wave function Ψ_n is such that:

$$\boldsymbol{\hat{H}\Psi_n = E_n\Psi_n}$$

$$-\frac{\hbar^2}{2m}\frac{d^2\Psi_n}{dx^2} + \boldsymbol{V(x)\Psi_n = E_n\Psi_n}$$

$$\frac{d^2\Psi_n}{dx^2} + \frac{2m}{\hbar^2}\left(\boldsymbol{E_n - V(x)}\right)\boldsymbol{\Psi_n = 0}$$

After solving the above equation one can evaluate the Eigen values of the Hamiltonian operator**.**

4. Particle in a one dimensional infinite potential well:

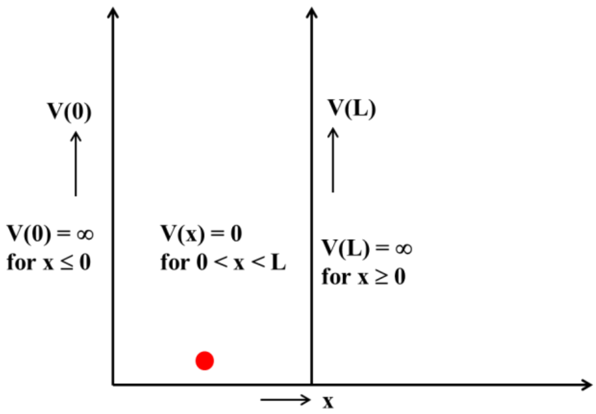


Fig. 1. A square potential well with infinitely high barriers at each end corresponds to a box with infinitely hard walls

Suppose a particle of mass m, is trapped in an infinite potential well whose boundaries are defined as

$$V(x) = 0 \text{ for } 0 < x < L$$

$$V(x) = \infty \text{ for } x \leq 0 \text{ and } x \geq L$$

Please note that infinitely high potential well correspond infinitely hard walls which mean that particle is trapped in a one dimensional box having infinitely hard walls.

The trapped particle does not lose energy when it collides with the walls of the box.

We are supposed to find out the particle wave functionΨ, and energy Eigen values.

Suppose the particle's associated wave function is $\boldsymbol{\Psi}$ and energy Eigen values is E_n.

Now writing the time independent Schrodinger wave equation for the particle in the range $0 < x < L$.

$$\frac{d^2\Psi_n}{dx^2} + \frac{2m}{\hbar^2}\left(E_n - V(x)\right)\Psi_n = 0$$

$$\Longrightarrow \frac{d^2\Psi_n}{dx^2} + \frac{2mE_n}{\hbar^2}\Psi_n = 0 \text{ ....(1)}$$

Suppose $\boldsymbol{k} = \sqrt{\frac{2mE_n}{\hbar^2}}$ then equation (1) can be written as

$$\frac{d^2\Psi_n}{dx^2} + k^2\Psi_n = 0 \text{ ....(2)}$$

Equation (2) is the differential equation of second order and first degree. In order to solve this equation, first we will write the auxiliary equation of Equation (2) and find its roots.

Auxiliary equation of equation (2),

$$D^2 + k^2 = 0 \quad \text{.....................(3)}$$

Note: *In order to write auxiliary equation of any differential equation we use to represent $D \equiv \frac{d}{dx}$ and remove the wave function . Then we write rest of the quantities as shown in equation (3) and find the roots.*

Roots of Equation (3), is $D = \pm ik$

Thus the solution of Equation (2) can be written as:

$$\Psi_n = A\ sin\ kx + B\ cos\ kx \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(4)$$

Where A and B are constants and can be determined applying the boundary conditions. As you know the particle is strictly bounded in the region 0 < x <L, so the particle wave function will be zero at $x = 0\ and\ L$

Thus at $x = 0$ $\Psi_n = 0$ thus from Equation (4)

$$0 = A \sin 0 + B\ cos\ 0 \Longrightarrow \mathrm{B} = 0$$

Thus the equation (4) can be written as

$$\Psi_n = A\ sin\ kx \ldots\ldots\ldots\ldots\ldots\ldots\ldots..\ (5)$$

Again at $x = L$ $\Psi_n = 0$, therefore equation (5) gives

$$\Psi_n = A\ sin\ kL = 0$$

$$sin\ kL = 0$$

$$kL = n\pi \ldots\ldots\ldots\ldots(6)$$

$$(kL)^2 = (n\pi)^2$$

$$\frac{\boldsymbol{2mE_nL^2}}{\hbar^2} = n^2\pi^2$$

$$\boldsymbol{E_n = \frac{\hbar^2 n^2 \pi^2}{2mL^2} \ldots\ldots\ldots\ldots\ldots\ldots(7)}$$

Equation (7) gives the energy Eigen values of the particle where n = 1,2,3,……..

Substituting the value of $k = \frac{n\pi}{L}$ from equation (6) in equation (5) we have

$$\Psi_n = A\ sin\ \frac{n\pi x}{L} \ldots\ldots\ldots\ldots\ldots(8)$$

Applying normalization condition in order to find A

$$\int_{-\infty}^{\infty} \Psi_n{}^* \ \Psi_n \ \mathrm{dx} = 1$$

$$\int_0^L \mathrm{A}^2\ sin^2\ \frac{n\pi x}{L}\ \mathrm{dx} = 1$$

$$\int_0^L \frac{A^2}{2} 2\, sin^2 \frac{n\pi x}{L}\, dx = 1$$

$$\frac{A^2}{2} \int_0^L \left(1 - \cos\frac{2n\pi x}{L}\right) dx = 1$$

$$\frac{A^2}{2} \left[x - \frac{L}{2n\pi} sin \frac{2n\pi x}{L}\right]_0^L = 1$$

$$A = \sqrt{\frac{2}{L}}$$

Thus equation (5) can be written as

$$\Psi_n = \sqrt{\frac{2}{L}}\, sin\, \frac{n\pi x}{L} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (9)$$

Equation (9) is the Eigen function of the particle where n = 1,2,3,……..

Let us plot the wave function and probability density of the particle for different n values

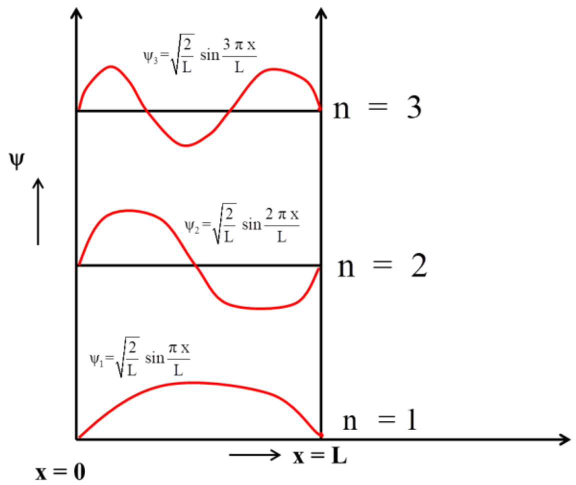


Fig. 2. Wave function plot

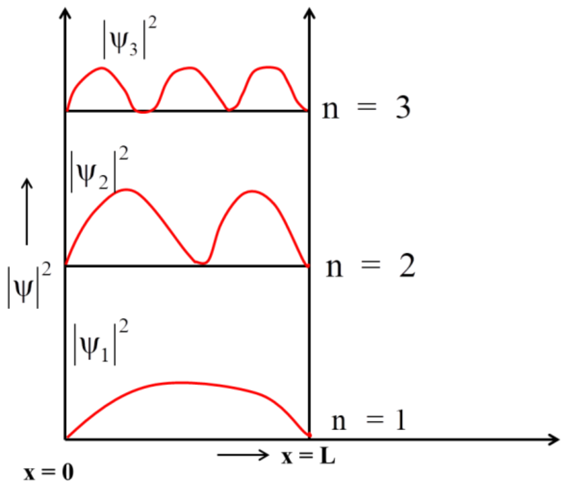


Fig. 3. Probability density of a particle

5. Particle in a one dimensional finite potential well:

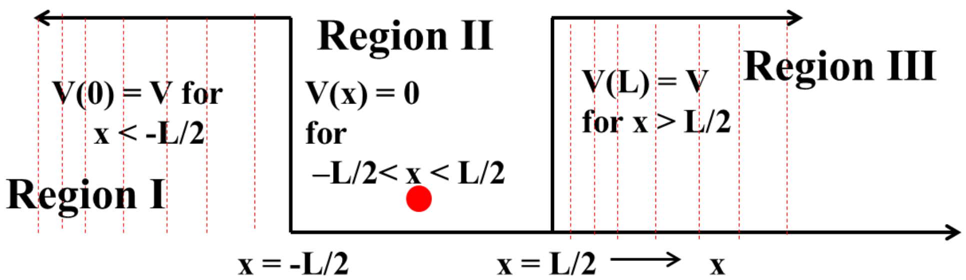


Fig. 1. A square potential well with finite height barriers at each end.

Previously discussed case of infinite potential well, is not possible in real world as the potential energies cannot be infinite. However the potential well with finite barrier does exist. Let us discuss the energy Eigen values and Eigen functions if particle is trapped in finite potential well.

Suppose a particle of mass m, is trapped in a symmetric finite potential well whose boundaries are defined as $V(x) = 0$ for $-\frac{L}{2} < x < \frac{L}{2}$

$V(x) = V$ for $x < -\frac{L}{2}$ and $x > \frac{L}{2}$

Suppose the particle energy E < V (height of the barrier). Classically the particle with energy E will be bounces back and forth in the finite potential well without entering the region I and III (refer to Fig. 1). But quantum mechanics says there is certain probability of penetration in the region I and III even though E < V.

(i) Writing the Schrodinger (Time independent) equation for **Region I**

Suppose the particle wave function in region I is Ψ_I

$$\frac{d^2\Psi_I}{dx^2} + \frac{2m}{\hbar^2}(E - V)\Psi_I = 0 \text{ ....(1)}$$

Suppose $k_1 = \sqrt{\frac{2m(V-E)}{\hbar^2}}$ then equation (1) can be written as

$$\frac{d^2\Psi_I}{dx^2} - k_1^2\Psi_I = 0 \text{.......................... (2)}$$

Equation (2) is the differential equation of second order and first degree. In order to solve this equation, first we will write the auxiliary equation of Equation (2) and find its roots.

Auxiliary equation of equation (2),

$$D^2 - k_1^2 = 0 \text{ .... (3)}$$

Note: *In order to write auxiliary equation of any differential equation we use to represent $D \equiv \frac{d}{dx}$ and remove the wave function. Then we write rest of the quantities as shown in equation (3) and find the roots.*

Roots of Equation (3), is $\quad D = \pm k_1$

Thus the solution of Equation (2) can be written as:

$$\Psi_I = Ae^{k_1 x} + Be^{-k_1 x} \text{......................... (4)}$$

Since property of wave function of the particle says that

Ψ_I must be finite everywhere and $\Psi_I \to 0\ as\ x \to -\infty$ for region I

Since equation (4) has two terms first term $Ae^{k_1 x}$ satisfy both the properties of wave function that is $\Psi_I \to 0\ as\ x \to -\infty$ and Ψ_I is finite everywhere. But second term of equation (4), $Be^{-k_1 x}$ does not satisfy above conditions and wave function $\Psi_I \to \infty\ as\ x \to -\infty$ so this term is useless and has to be dropped.

Thus the final solution of equation (2)

$$\boldsymbol{\Psi_I = Ae^{k_1 x}} \text{.................. } \mathbf{(5)}$$

(ii) Writing the Schrodinger (Time independent) equation for **Region II**

$$\boldsymbol{\frac{d^2\Psi_{II}}{dx^2} + \frac{2mE}{\hbar^2}\Psi_{II} = 0} \text{......................}\mathbf{(6)}$$

Suppose $\boldsymbol{k_2} = \sqrt{\frac{\boldsymbol{2mE}}{\hbar^2}}$ then equation (6) can be written as

$$\boldsymbol{\frac{d^2\Psi_{II}}{dx^2} + k_2^2\Psi_{II} = 0} \text{...................}\mathbf{(7)}$$

Solution of equation (7) with antisymmetric wave function

$$\Psi_{II} = B\ sin\ k_2 x \text{........................... (8)}$$

Solution of equation (7) with symmetric wave function

$$\Psi_{II} = C\ cos\ k_2 x \text{...................... (9)}$$

(iii) Writing the Schrodinger (Time independent) equation for **Region III**

$$\boldsymbol{\frac{d^2\Psi_{III}}{dx^2} - k_1^2\Psi_{III} = 0} \text{.......................... } \mathbf{(10)}$$

Solution of equation (10) can be written as

$$\boldsymbol{\Psi_{III} = De^{-\,k_1 x}} \text{.................. } \mathbf{(11)}$$

(Antisymmetric case that is wave function given by Eq. (8)) Applying the boundary conditions at $x = -\frac{L}{2}$,

$$\boldsymbol{\Psi_I = \Psi_{II}}$$

$$\boldsymbol{Ae^{\frac{-k_1 L}{2}}} = -B\ sin\ \frac{k_2 L}{2} \text{..................(12)}$$

Further first derivative of the wave function must be continuous at $x = -\frac{L}{2}$

$$\frac{\mathrm{d}\Psi_I}{dx} = \frac{\mathrm{d}\Psi_{II}}{dx}$$

$$\boldsymbol{Ak_1 e}^{-\frac{k_1 L}{2}} = -B\; k_2 cos\; \frac{k_2 L}{2} \text{.........(13)}$$

Dividing Eq. (13) by (12) we have

$$\cot\frac{k_2 L}{2} = \frac{k_1}{k_2} = \sqrt{\frac{(V-E)}{E}} \text{...................... (14)}$$

The roots of the transcendental Eq (14) will give the discrete values of E for antisymmetric case.

For $V \rightarrow \infty$ the RHS of equation (14) will become

$$\cot\frac{k_2 L}{2} = \infty = \cot n\pi$$

$$\Rightarrow \frac{k_2 L}{2} = n\pi$$

$$\Rightarrow \left(\frac{k_2 L}{2}\right)^2 = (n\pi)^2$$

$$\Longrightarrow \frac{\boldsymbol{2mEL^2}}{\boldsymbol{4\hbar^2}} = (n\pi)^2$$

$$\Longrightarrow \boldsymbol{E} = \frac{(2n\pi)^2\hbar^2}{2m\boldsymbol{L^2}}$$ Where n =1, 2, 3, 4…….

(Symmetric case that is wave function given by Eq. (9)) Applying the boundary conditions at $x = -\frac{L}{2}$,

$$\boldsymbol{\Psi_I} = \boldsymbol{\Psi_{II}}$$

$$\boldsymbol{Ae}^{\frac{-k_1 L}{2}} = \boldsymbol{C}\; cos\; \frac{k_2 L}{2} \text{…………….. (15)}$$

Further first derivative of the wave function must be continuous at $x = -\frac{L}{2}$

$$\frac{\mathrm{d}\Psi_I}{dx} = \frac{\mathrm{d}\Psi_{II}}{dx}$$

$$\boldsymbol{Ak_1 e}^{-\frac{k_1 L}{2}} = \boldsymbol{C}\; k_2 sin\; \frac{k_2 L}{2} \text{……… (16)}$$

Dividing Eq. (16) by (15) we have

$$\tan\frac{k_2 L}{2} = \frac{k_1}{k_2} = \sqrt{\frac{(V-E)}{E}} \text{...................... (17)}$$

The roots of the transcendental Eq (17) will give the discrete values of E for symmetric case.

For $V \rightarrow \infty$ the RHS of equation (17) will become

$$\tan\frac{k_2 L}{2} = \infty = \tan\left(n + \frac{1}{2}\right)\pi$$

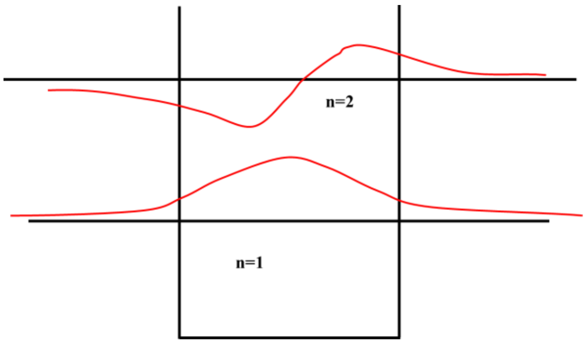
$$\Rightarrow \frac{k_2 L}{2} = \left(n + \frac{1}{2}\right)\pi$$

$$\Rightarrow \left(\frac{k_2 L}{2}\right)^2 = \left(n + \frac{1}{2}\right)^2 \pi^2$$

$$\Longrightarrow \frac{\mathbf{2}\boldsymbol{mEL}^{\mathbf{2}}}{\mathbf{4}\boldsymbol{\hbar}^{\mathbf{2}}} = \left(n + \frac{1}{2}\right)^2 \pi^2$$

$$\Longrightarrow \boldsymbol{E} = \frac{4\left(n+\frac{1}{2}\right)^2 \pi^2 \hbar^{\mathbf{2}}}{2m\boldsymbol{L}^{\mathbf{2}}}$$

$$\Longrightarrow \boldsymbol{E} = \frac{(2n+1)^2 \pi^2 \hbar^{\mathbf{2}}}{2m\boldsymbol{L}^{\mathbf{2}}}$$ Where $n = 0,1, 2, 3, 4\ldots\ldots$



Wave function of the particle in finite symmetric potential well